Relatório 9 - Prática: Git e Github para Iniciantes (III)

Gabriel Oliveira Santos

1. Introdução

O objetivo desta aula foi aprender sobre Git e Github de uma maneira leve. As aulas contemplavam desde o básico até comandos que eu não conhecia, mesmo sendo usuário da ferramenta. A principal vantagem de aprender é que hoje, para versionamento de código, são as mais utilizadas.

2. Desenvolvimento



A muitos anos, pessoas vem sofrendo com uma principal questão dentro do contexto de versão dos códigos: Organização. O versionamento de código nos dias de hoje é mais do que essencial, sendo primordial para projetos em que um time está fazendo atualizações o tempo todo.

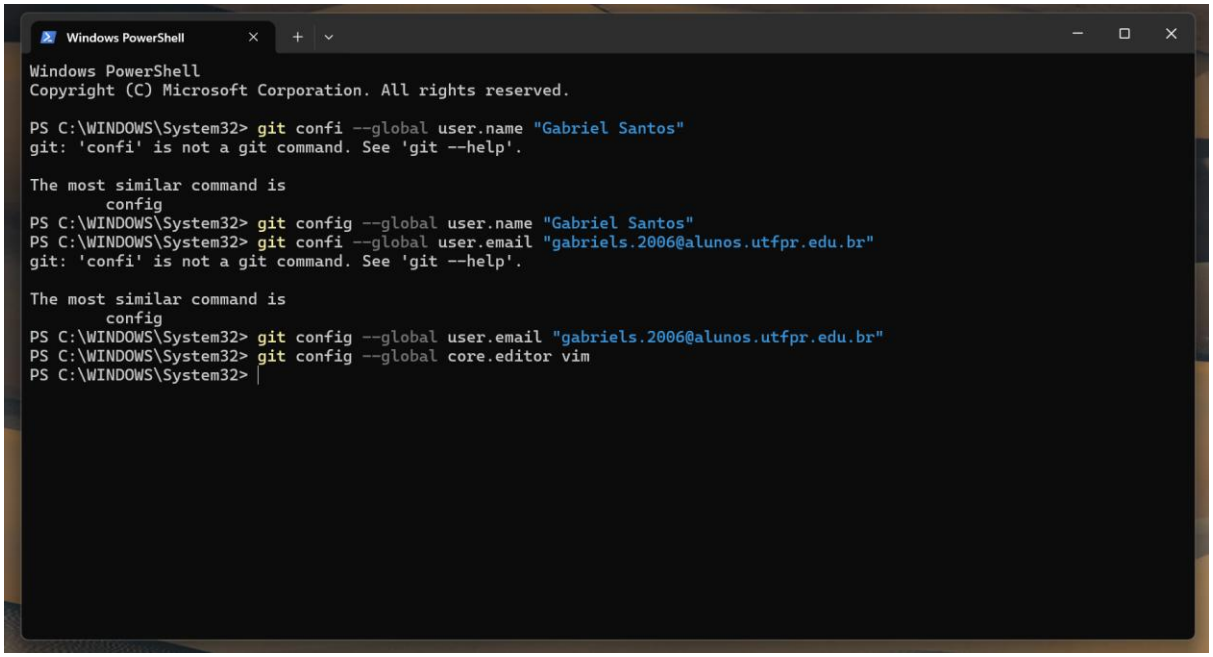
De certa forma, o git e github revolucionaram essa ideia. Sua forma, que em comparação aos outros sistemas é diferente, guarda o estado dos arquivos versão a versão, como uma foto. Isso permite ir e vir entre as diferentes versões. Outros sistemas guardam as diferenças dos arquivos.

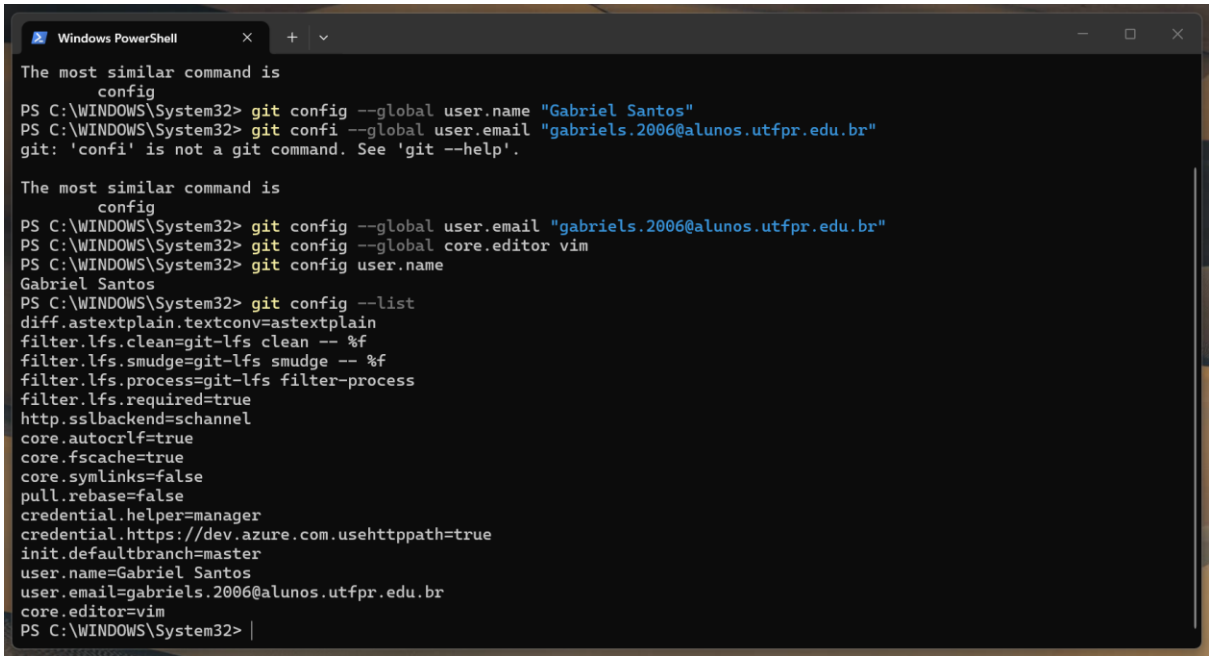
Sua história remonta ao famigerado Linux, onde em uma confusão empresarial, acabou resultando na criação do Git pelo criador do Linux.

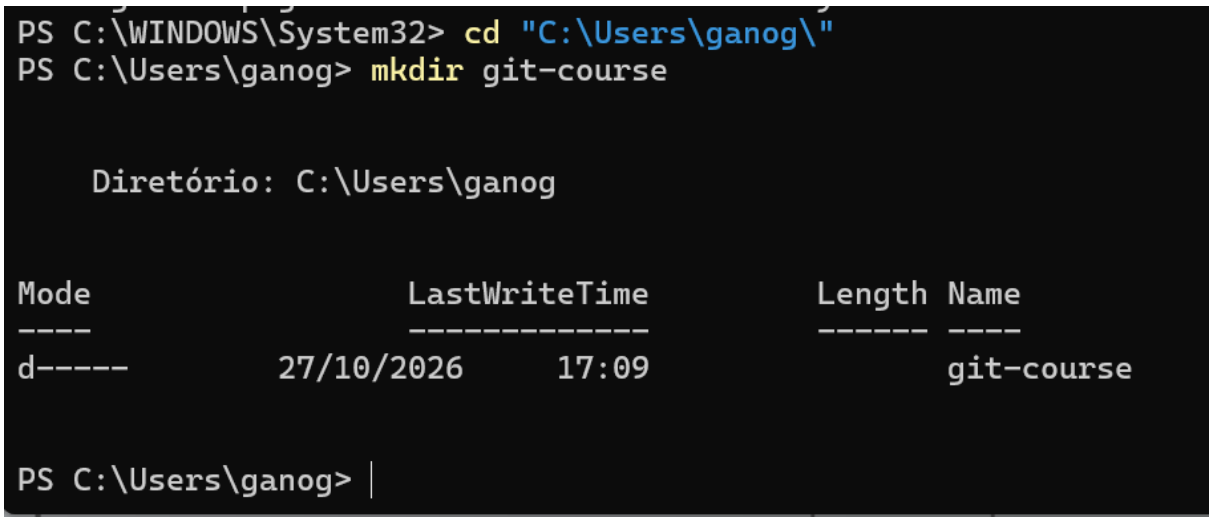
O Github é apenas um local na web em que armazena os projetos versionados pelo Git, sem o Git (controle) não existiria o GitHub (repositório remoto).

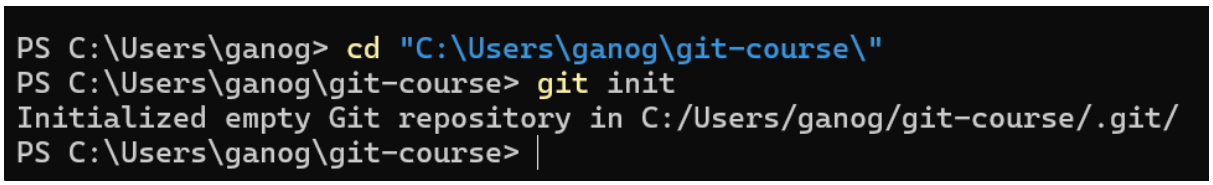


- Git Config sys, user, project









Aqui alterno para o GIT BASH pois estou no Windows 11 e não é necessário baixar o VIM para usar pelo powershell por ter isso já instalado

comandos

git init; git add;
git commit;
git status; git push;
git clone;
git branch; git merge
git rebase;

```
MINGW64:/c/Users/ganog/git-course

ganog@Dell-Gabriel MINGW64 ~
$ cd "C:\Users\ganog\git-course"

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ vim Readme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ vim Readme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$
```

```
MINGW64:/c/Users/ganog/git-course
# Github

Arquivo da aula de Git e GitHub para iniciantes
~
~
~
~
~
~
~
~
```

```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git add Readdme.md
warning: in the working copy of 'Readdme.md', LF will be replaced by CRLF the next time Git touches it

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   Readdme.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md


ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ vim Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master

No commits yet

Changes to be committed:
  (use "git rm --cached <file>..." to unstage)
        new file:   Readdme.md

Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Readdme.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md


ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git add Readdme.md
warning: in the working copy of 'Readdme.md', LF will be replaced by CRLF the next time Git touches it

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git commit -m "Add Readme.md e Readdme.md"
[master (root-commit) bcb6af6] Add Readme.md e Readdme.md
 1 file changed, 2 insertions(+)
 create mode 100644 Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$
```

```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ vim Readme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md

nothing added to commit but untracked files present (use "git add" to track)

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git commit -m "Vai dar erro"
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md

nothing added to commit but untracked files present (use "git add" to track)

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ vim Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Readdme.md

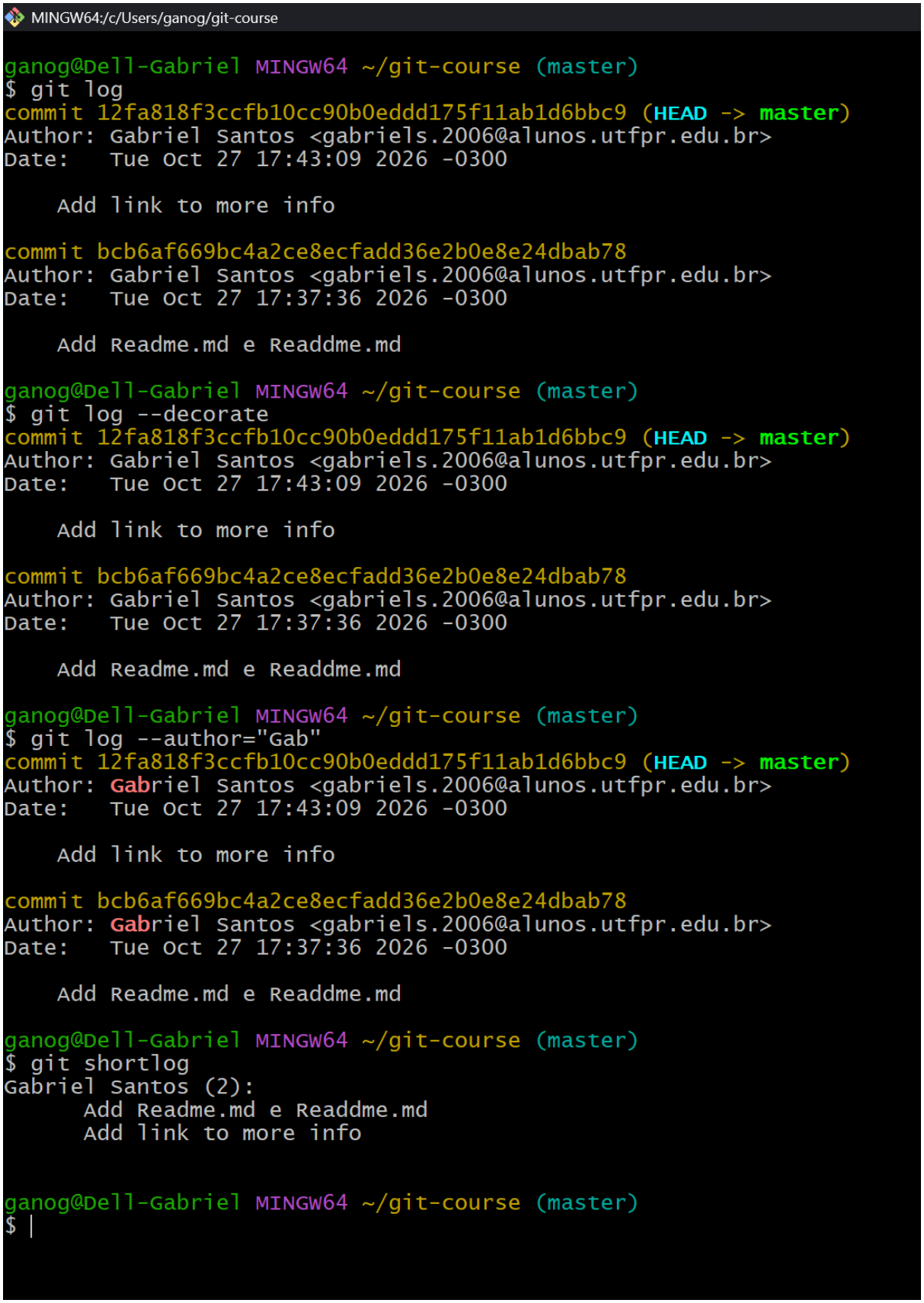
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md

no changes added to commit (use "git add" and/or "git commit -a")

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$
```

O Git tem algumas marcações importantes como Untracked, Unmodified, Modified e Staged. Em ordem, o arquivo acaba de ser colocado, depois altera o estado para o segundo, se editado, vai para o terceiro e por fim no staged, onde cria uma versão nova, voltando ao unmodified.

Histórico de commits até o momento pelo usuário Gabriel (Eu)


MINGW64:/c/Users/ganog/git-course
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git log
commit 12fa818f3ccfb10cc90b0eddd175f11ab1d6bbc9 (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 17:43:09 2026 -0300
Add link to more info
commit bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 17:37:36 2026 -0300
Add Readme.md e Readdme.md
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git log --decorate
commit 12fa818f3ccfb10cc90b0eddd175f11ab1d6bbc9 (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 17:43:09 2026 -0300
Add link to more info
commit bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 17:37:36 2026 -0300
Add Readme.md e Readdme.md
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git log --author="Gab"
commit 12fa818f3ccfb10cc90b0eddd175f11ab1d6bbc9 (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 17:43:09 2026 -0300
Add link to more info
commit bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 17:37:36 2026 -0300
Add Readme.md e Readdme.md
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git shortlog
Gabriel Santos (2):
Add Readme.md e Readdme.md
Add link to more info
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$

```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git shortlog -sn
     2  Gabriel Santos

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git log --graph
* commit 12fa818f3ccfb10cc90b0eddd175f11ab1d6bbc9 (HEAD -> master)
| Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
| Date:   Tue Oct 27 17:43:09 2026 -0300
|
|     Add link to more info
|
* commit bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78
  Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
  Date:   Tue Oct 27 17:37:36 2026 -0300

      Add Readme.md e Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ ^C

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git show 12fa818f3ccfb10cc90b0eddd175f11ab1d6bbc9
commit 12fa818f3ccfb10cc90b0eddd175f11ab1d6bbc9 (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:43:09 2026 -0300

    Add link to more info

diff --git a/Readdme.md b/Readdme.md
index 642cf9d..becfa9c 100644
--- a/Readdme.md
+++ b/Readdme.md
@@ -1,2 +1,4 @@
 # Git Course
 Este é um repositório para mostrar o como o git funciona
+
+Saiba mais em : [willianjusten.com.br](http://willianjusten.com.br)

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$
```

```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md

nothing added to commit but untracked files present (use "git add" to track)

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ vim Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git diff
warning: in the working copy of 'Readdme.md', LF will be replaced by CRLF the next time Git touches it
diff --git a/Readdme.md b/Readdme.md
index becfa9c..53134a4 100644
--- a/Readdme.md
+++ b/Readdme.md
@@ -2,3 +2,5 @@
 Este é um repositório para mostrar o como o git funciona

 Saiba mais em : [willianjusten.com.br](http://willianjusten.com.br)
+
+Gostou do curso? Quer mais? Ajude com uma doação, até um café é valido =)

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git diff -name-only
fatal: 'ame-only': not an integer

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git diff --name-only
warning: in the working copy of 'Readdme.md', LF will be replaced by CRLF the next time Git touches it
Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git commit-am "Edit Readdme"
git: 'commit-am' is not a git command. See 'git --help'.

The most similar command is
        commit-graph

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git commit -am "Edit Readdme"
warning: in the working copy of 'Readdme.md', LF will be replaced by CRLF the next time Git touches it
[master ed721cb] Edit Readdme
 1 file changed, 2 insertions(+)

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$
```

```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git show
commit ed721cb1293fed10d676c187dc32347b5df7a44c (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:49:02 2026 -0300

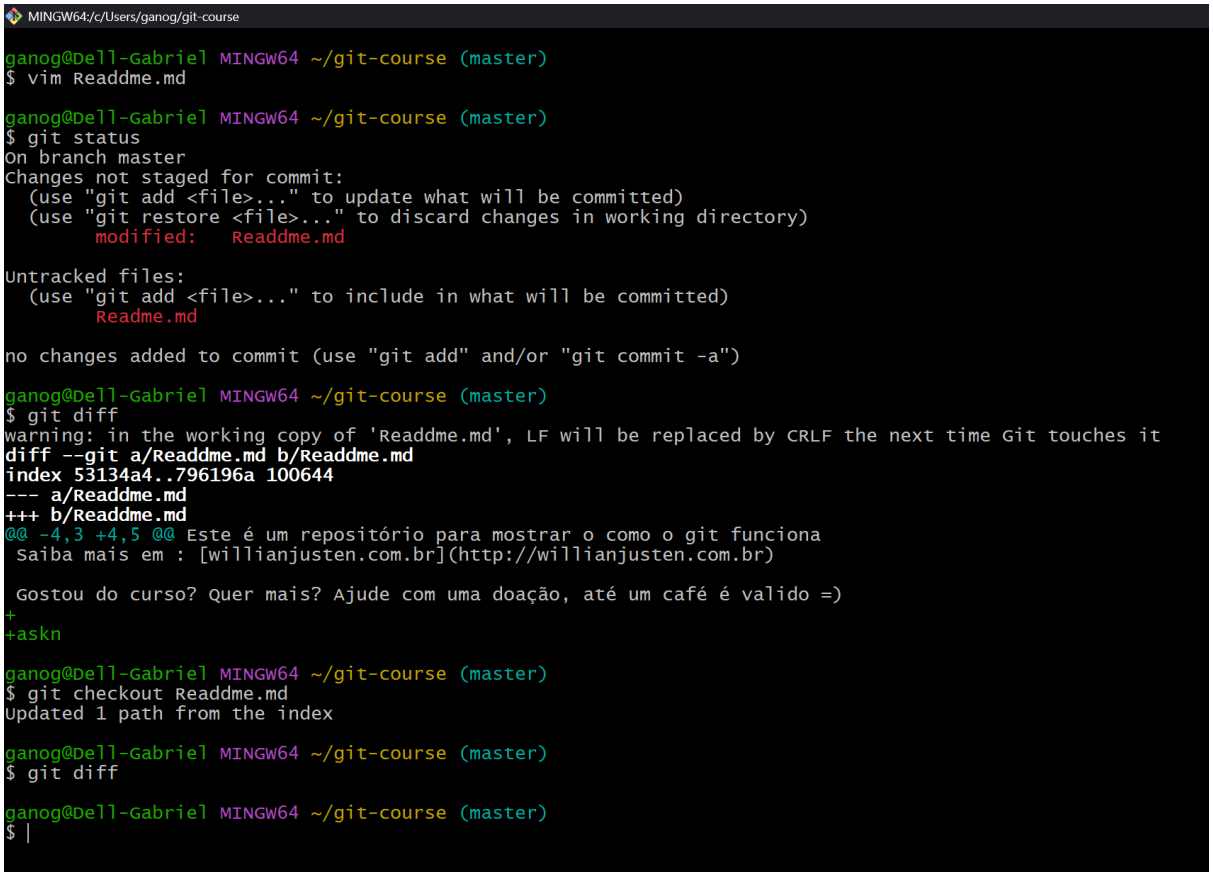
    Edit Readdme

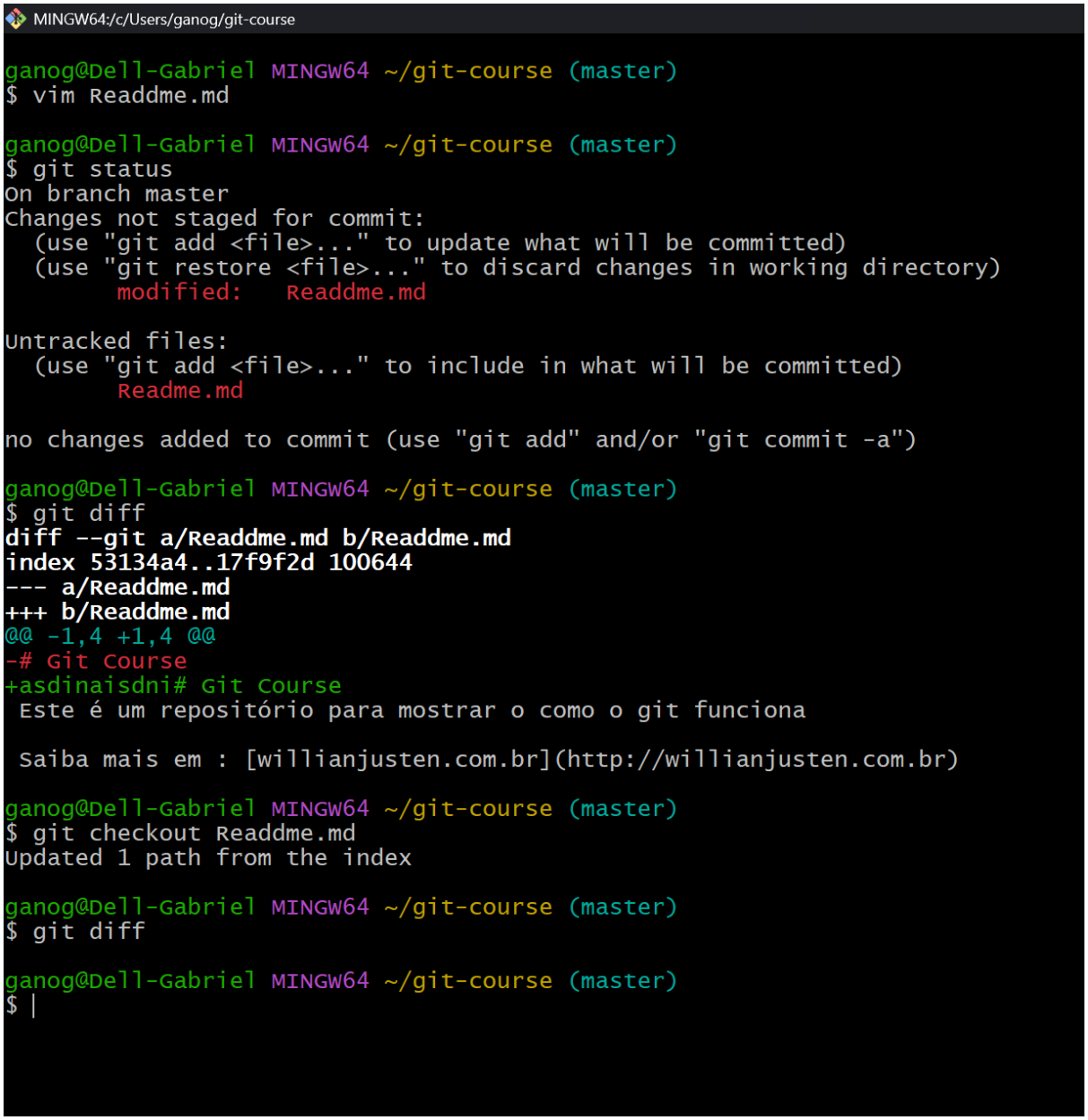
diff --git a/Readdme.md b/Readdme.md
index becfa9c..53134a4 100644
--- a/Readdme.md
+++ b/Readdme.md
@@ -2,3 +2,5 @@
 Este é um repositório para mostrar o como o git funciona

 Saiba mais em : [willianjusten.com.br](http://willianjusten.com.br)
+
+Gostou do curso? Quer mais? Ajude com uma doação, até um café é valido =)

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$
```

- Git diff pega a diferença entre um commit e outro, útil para comparar código.




MINGW64:/c/Users/ganog/git-course
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ vim Readdme.md
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master
Changes not staged for commit:
(use "git add <file>..." to update what will be committed)
(use "git restore <file>..." to discard changes in working directory)
modified: Readdme.md
Untracked files:
(use "git add <file>..." to include in what will be committed)
Readme.md
no changes added to commit (use "git add" and/or "git commit -a")
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git diff
diff --git a/Readdme.md b/Readdme.md
index 53134a4..17f9f2d 100644
--- a/Readdme.md
+++ b/Readdme.md
@@ -1,4 +1,4 @@
-# Git Course
+asdinaisdni# Git Course
Este é um repositório para mostrar o como o git funciona
Saiba mais em : willianjusten.com.br
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git checkout Readdme.md
Updated 1 path from the index
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git diff
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$

```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ vim Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git add Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   Readdme.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md


ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git diff

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
```

```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ vim Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git add Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   Readdme.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md


ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git diff

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git reset HEAD Readdme.md
Unstaged changes after reset:
M       Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git checkout Readdme.md
Updated 1 path from the index
```

```
MINGW64:/c/Users/ganog/git-course

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ vim Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   Readdme.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md

no changes added to commit (use "git add" and/or "git commit -a")

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git commit -am "asndioani"
[master f4c024a] asndioani
 1 file changed, 2 insertions(+)

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git log
commit f4c024a7fd29fe85af3282aa62268a3fd0886812 (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:58:42 2026 -0300

    asndioani

commit ed721cb1293fed10d676c187dc32347b5df7a44c
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:49:02 2026 -0300

    Edit Readdme

commit 12fa818f3ccfb10cc90b0eddd175f11ab1d6bbc9
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:43:09 2026 -0300

    Add link to more info

commit bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:37:36 2026 -0300

    Add Readme.md e Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git reset --soft --mixed --hard
HEAD is now at f4c024a asndioani
```

```
MINGW64:/c/Users/ganog/git-course

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git log
commit f4c024a7fd29fe85af3282aa62268a3fd0886812 (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:58:42 2026 -0300

    asndioani

commit ed721cb1293fed10d676c187dc32347b5df7a44c
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:49:02 2026 -0300

    Edit Readdme

commit 12fa818f3ccfb10cc90b0eddd175f11ab1d6bbc9
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:43:09 2026 -0300

    Add link to more info

commit bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:37:36 2026 -0300

    Add Readme.md e Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git reset --soft bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78




ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   Readdme.md

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md


ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git commit -m "kkkkk"
[master a5fcc0c] kkkkk
 1 file changed, 6 insertions(+)

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
```

```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git reset --mixed bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78
Unstaged changes after reset:
M       Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git log
commit bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78 (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:37:36 2026 -0300

    Add Readme.md e Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$
```

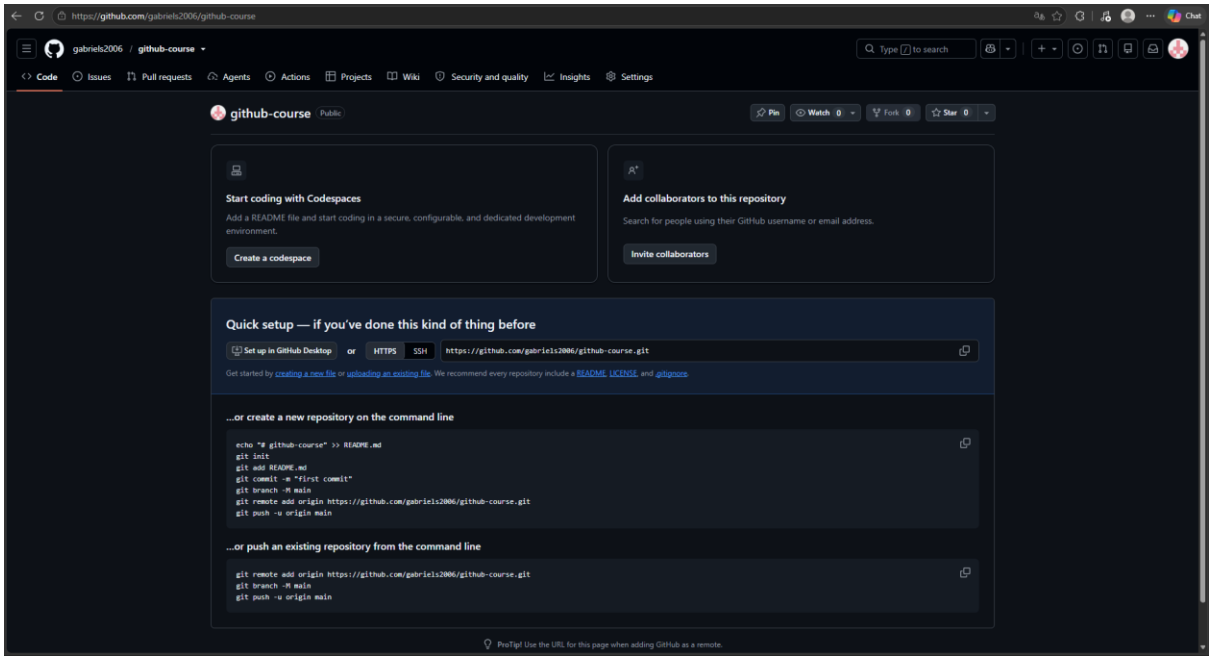
```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git reset --hard bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78
HEAD is now at bcb6af6 Add Readme.md e Readdme.md

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git log
commit bcb6af669bc4a2ce8ecfadd36e2b0e8e24dbab78 (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date:   Tue Oct 27 17:37:36 2026 -0300

    Add Readme.md e Readdme.md
ggi
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        Readme.md

nothing added to commit but untracked files present (use "git add" to track)
```

- Criando repositório que é público: https://github.com/gabriels2006/github-course



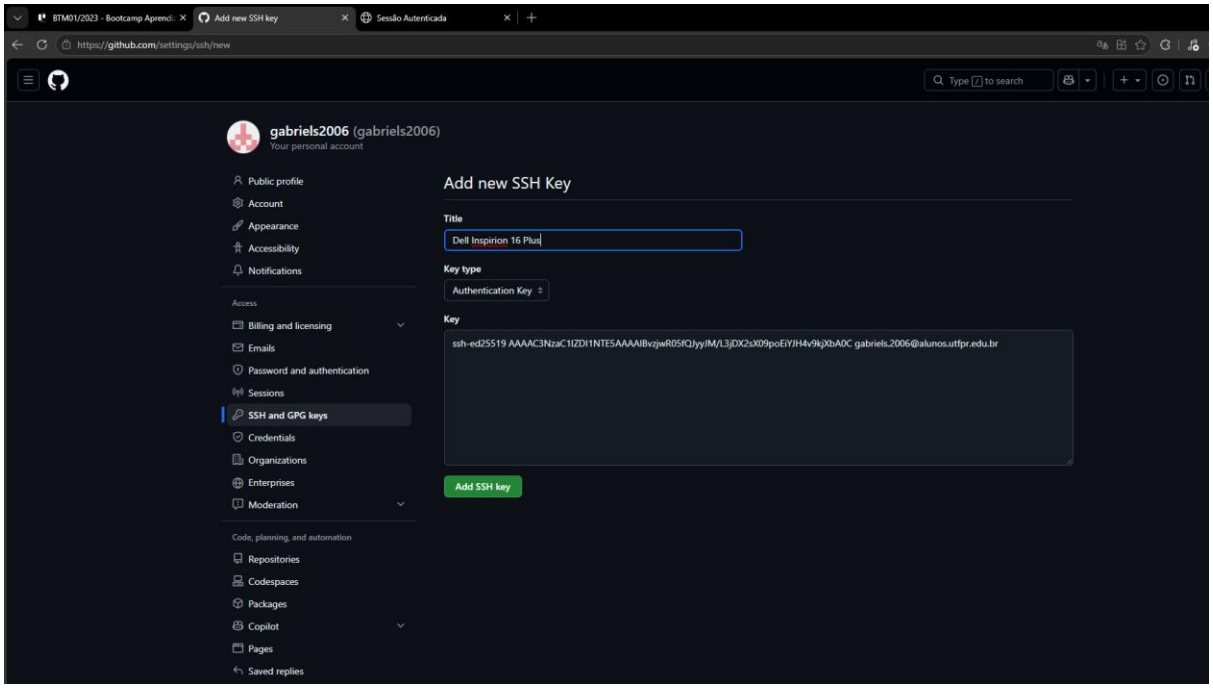

```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ ssh-keygen -t ed25519 -C "gabriels.2006@alunos.utfpr.edu.br"
Generating public/private ed25519 key pair.
Enter file in which to save the key (/c/Users/ganog/.ssh/id_ed25519):
Created directory '/c/Users/ganog/.ssh'.
Enter passphrase for "/c/Users/ganog/.ssh/id_ed25519" (empty for no passphrase):
Enter same passphrase again:
Your identification has been saved in /c/Users/ganog/.ssh/id_ed25519
Your public key has been saved in /c/Users/ganog/.ssh/id_ed25519.pub
The key fingerprint is:
SHA256:65uDMUTptUc3klt5QDeXvNg/SLwxw659tSWlwT0OBOs gabriels.2006@alunos.utfpr.edu.br
The key's randomart image is:
+--[ED25519 256]--+
|        .   oo+*.o|
|      o . + =.oB  |
|     o . o =o=+.o|
|      o . o .OO+o|
|     .  S.  oE*+o|
|      o  .   +o.+|
|       +.   o  .=|
|      .... . . o |
|        +o    .  |
+----[SHA256]-----+

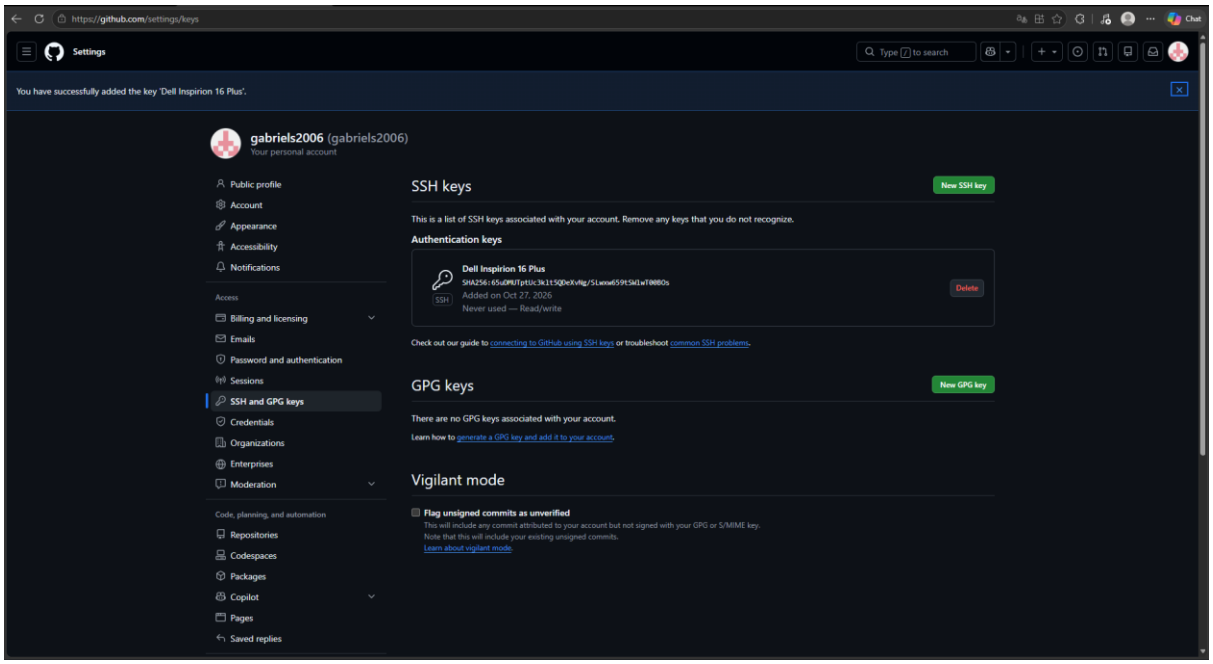
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ cd ~/.
./                .claude/            .cursor/            .jdks/              .packettracer       .vscode/
../               .claude.json        .gitconfig          .junie/             .python_history     .vscode-shared/
.VirtualBox/      .claude.json.backup .ipynb_checkpoints/ .jupyter/           .ssh/
.agents/          .copilot/           .ipython/           .lesshst            .viminfo

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ cd ~/.ssh

ganog@Dell-Gabriel MINGW64 ~/.ssh
$ ls
id_ed25519  id_ed25519.pub

ganog@Dell-Gabriel MINGW64 ~/.ssh
$
```


BTM01/2023 - Bootcamp Aprendi
Add new SSH key
Sessão Autenticada
https://github.com/settings/ssh/new
Type / to search
gabriels2006 (gabriels2006)
Your personal account
Public profile
Account
Appearance
Accessibility
Notifications
Access
Billing and licensing
Emails
Password and authentication
Sessions
SSH and GPG keys
Credentials
Organizations
Enterprises
Moderation
Code, planning, and automation
Repositories
Codespaces
Packages
Copilot
Pages
Saved replies
Add new SSH Key
Title
Dell Inspirion 16 Plus
Key type
Authentication Key
Key
ssh-ed25519 AAAAC3NzaC1lZDI1NTE5AAAAIBvzjwR05fQJyyJM/L3jDX2sX09poEiYJH4v9kjXbA0C gabriels.2006@alunos.utfpr.edu.br
Add SSH key


https://github.com/settings/keys
Settings
Type / to search
You have successfully added the key 'Dell Inspirion 16 Plus'.
gabriels2006 (gabriels2006)
Your personal account
Public profile
Account
Appearance
Accessibility
Notifications
Access
Billing and licensing
Emails
Password and authentication
Sessions
SSH and GPG keys
Credentials
Organizations
Enterprises
Moderation
Code, planning, and automation
Repositories
Codespaces
Packages
Copilot
Pages
Saved replies
SSH keys
New SSH key
This is a list of SSH keys associated with your account. Remove any keys that you do not recognize.
Authentication keys
Dell Inspirion 16 Plus
SHA256:65uDMUTptUc3k1tSQDeXvNg/5Lwxw659tSW1wT0BBOs
SSH
Added on Oct 27, 2026
Never used — Read/write
Delete
Check out our guide to connecting to GitHub using SSH keys or troubleshoot common SSH problems.
GPG keys
New GPG key
There are no GPG keys associated with your account.
Learn how to generate a GPG key and add it to your account.
Vigilant mode
Flag unsigned commits as unverified
This will include any commit attributed to your account but not signed with your GPG or S/MIME key.
Note that this will include your existing unsigned commits.
Learn about vigilant mode.

```
ganog@Dell-Gabriel MINGW64 ~/.ssh
$ cd "C:\Users\ganog\git-course"

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git remote add origin https://github.com/gabriels2006/github-course.git

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git remote -v
origin  https://github.com/gabriels2006/github-course.git (fetch)
origin  https://github.com/gabriels2006/github-course.git (push)

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git remote
origin

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$
```

```
ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git push -u origin main
error: src refspec main does not match any
error: failed to push some refs to 'https://github.com/gabriels2006/github-course.git'

ganog@Dell-Gabriel MINGW64 ~/git-course (master)
$ git branch -M main

ganog@Dell-Gabriel MINGW64 ~/git-course (main)
$ git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 303 bytes | 303.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/gabriels2006/github-course.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.

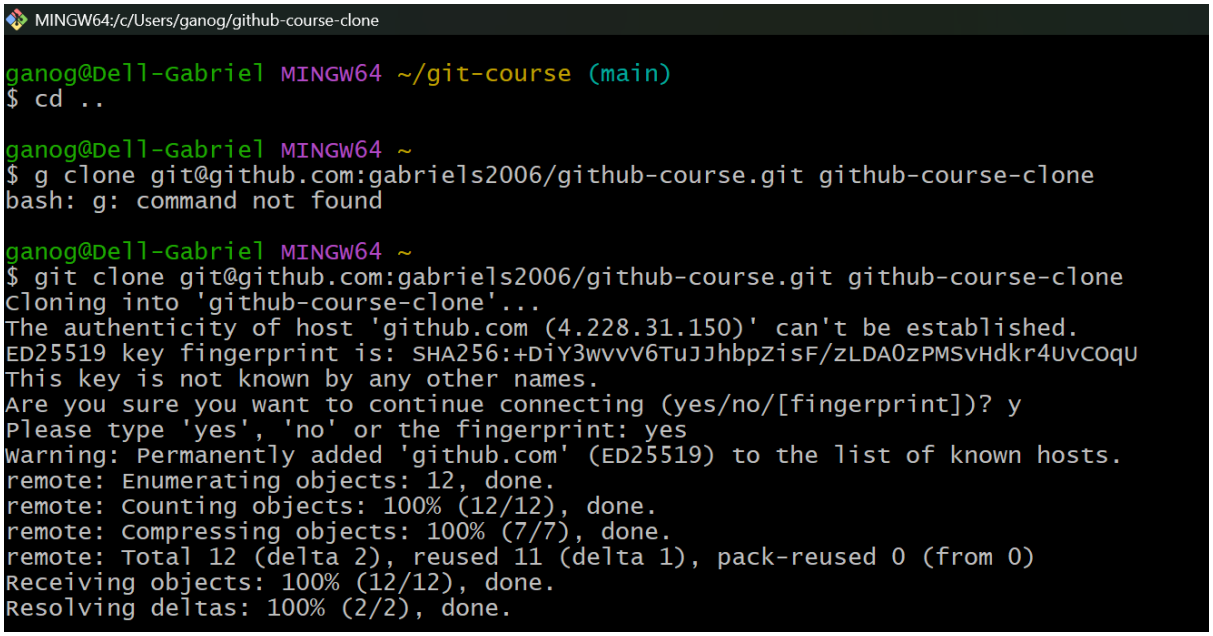
ganog@Dell-Gabriel MINGW64 ~/git-course (main)
$
```

```
ganog@Dell-Gabriel MINGW64 ~/git-course (main)
$ vim Readdme.md

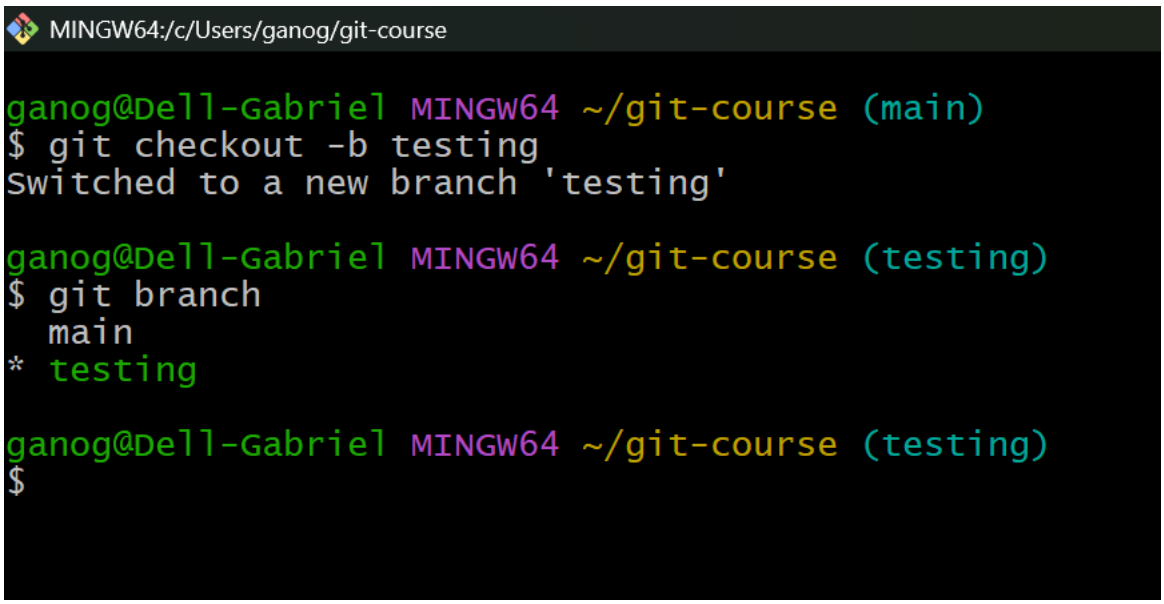
ganog@Dell-Gabriel MINGW64 ~/git-course (main)
$ git commit -am "Add Paypal button"
[main ee8482c] Add Paypal button
 1 file changed, 1 insertion(+), 1 deletion(-)

ganog@Dell-Gabriel MINGW64 ~/git-course (main)
$ git push origin main
Enumerating objects: 5, done.
Counting objects: 100% (5/5), done.
Delta compression using up to 16 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 298 bytes | 298.00 KiB/s, done.
Total 3 (delta 1), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (1/1), completed with 1 local object.
To https://github.com/gabriels2006/github-course.git
   5c3e88d..ee8482c  main -> main

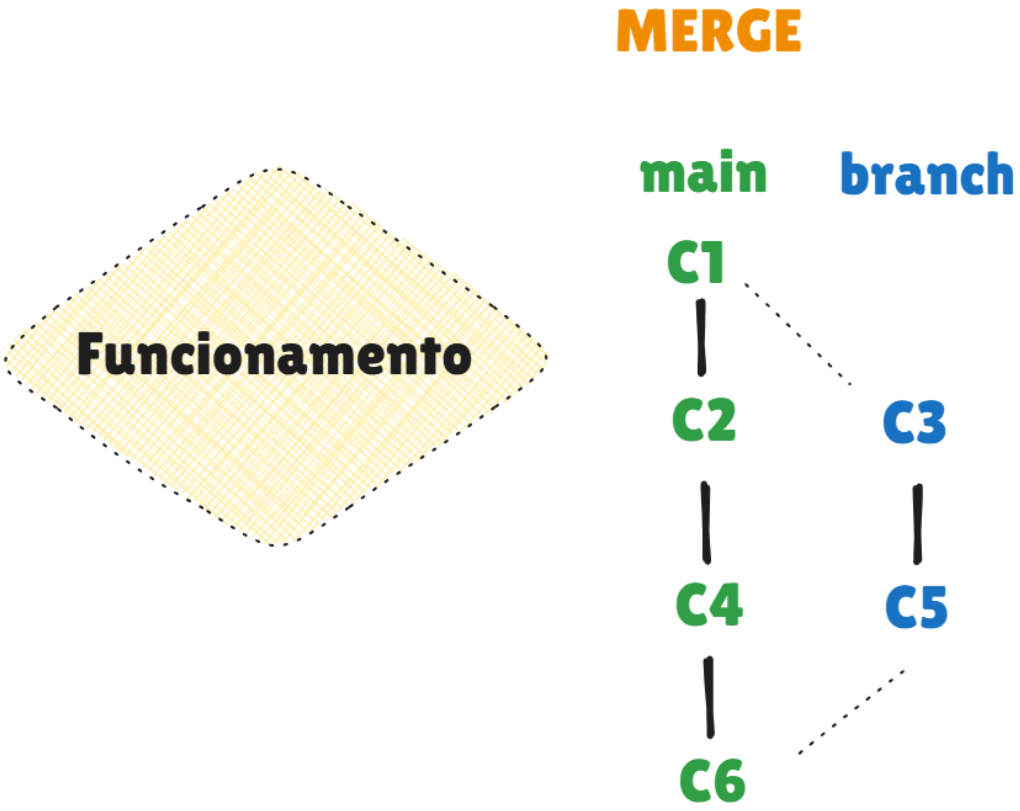
ganog@Dell-Gabriel MINGW64 ~/git-course (main)
$
```



O Branch é o ápice de todo o funcionamento do git, ele que separa os commits em partes. Serve como um ponteiro, que sai apontando para diferentes commits. Suas vantagens contempla a edição de arquivos enquanto tem outras pessoas mexendo no principal, na main. Ele é facilmente apagável, outro ponto que vale a pena, pois jogar na main direto pode resultar em conflitos, evitando assim conflitos.

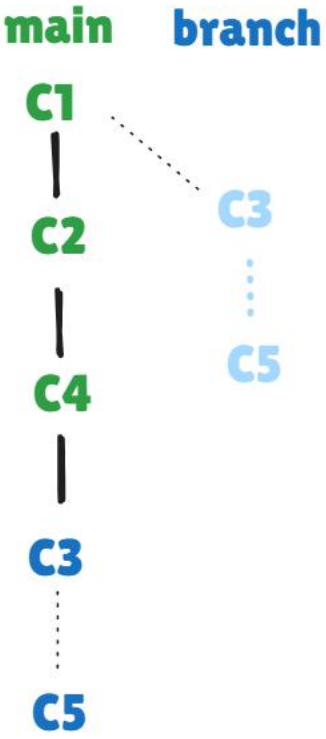


Existem alguns tipos de commit em uma Branch. Quando ela é nova, normalmente há muitas ramificações de atualizações rápidas, gerando um conflito de vários commits “fora de ordem”. O responsável por organizar essa ideia é o Merge, que acaba gerando um vértice/triângulo dos commits. Suas vantagens é que não estraga o histórico, entretanto, é necessário um commit extra pra juntar tudo.

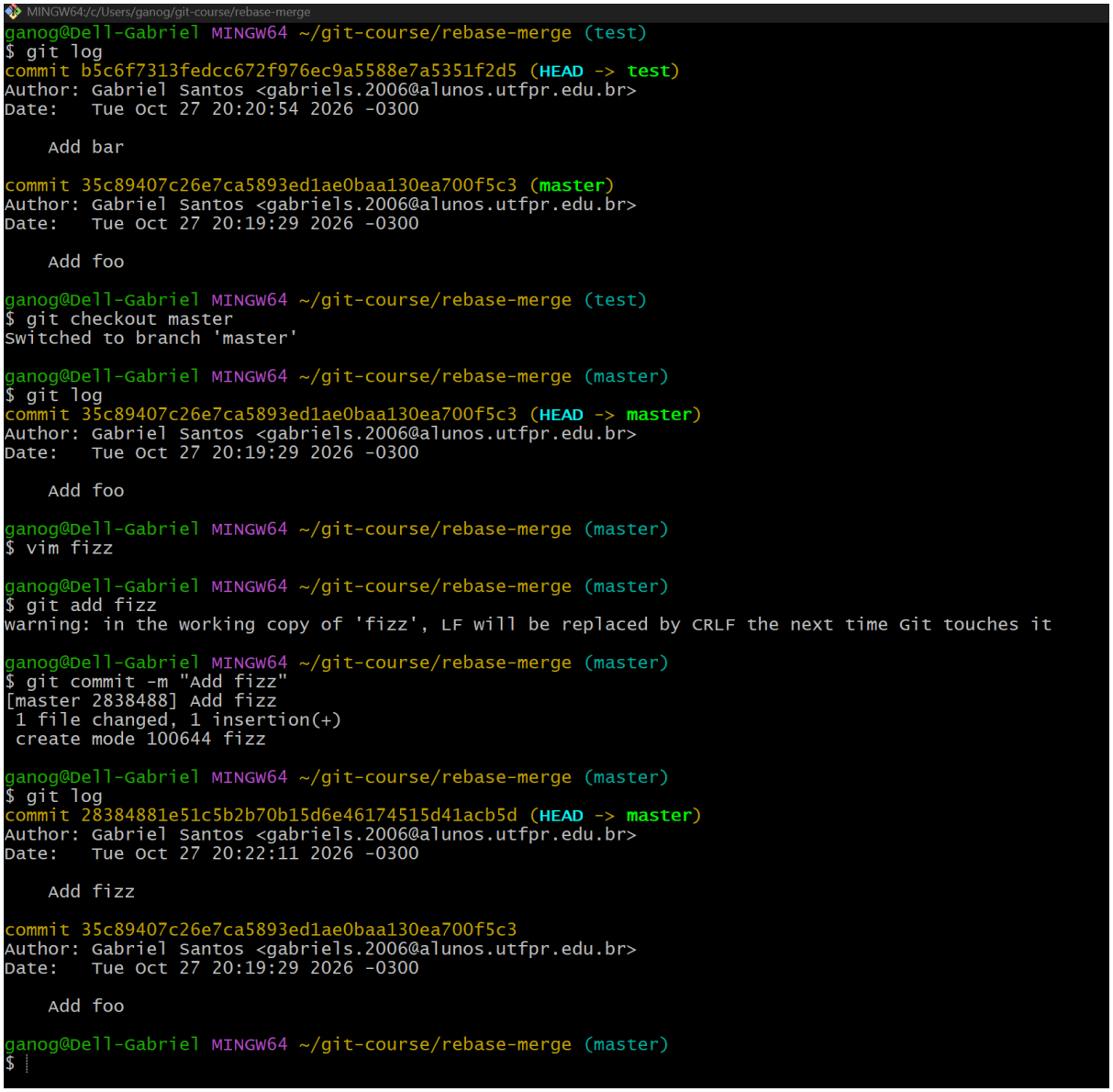


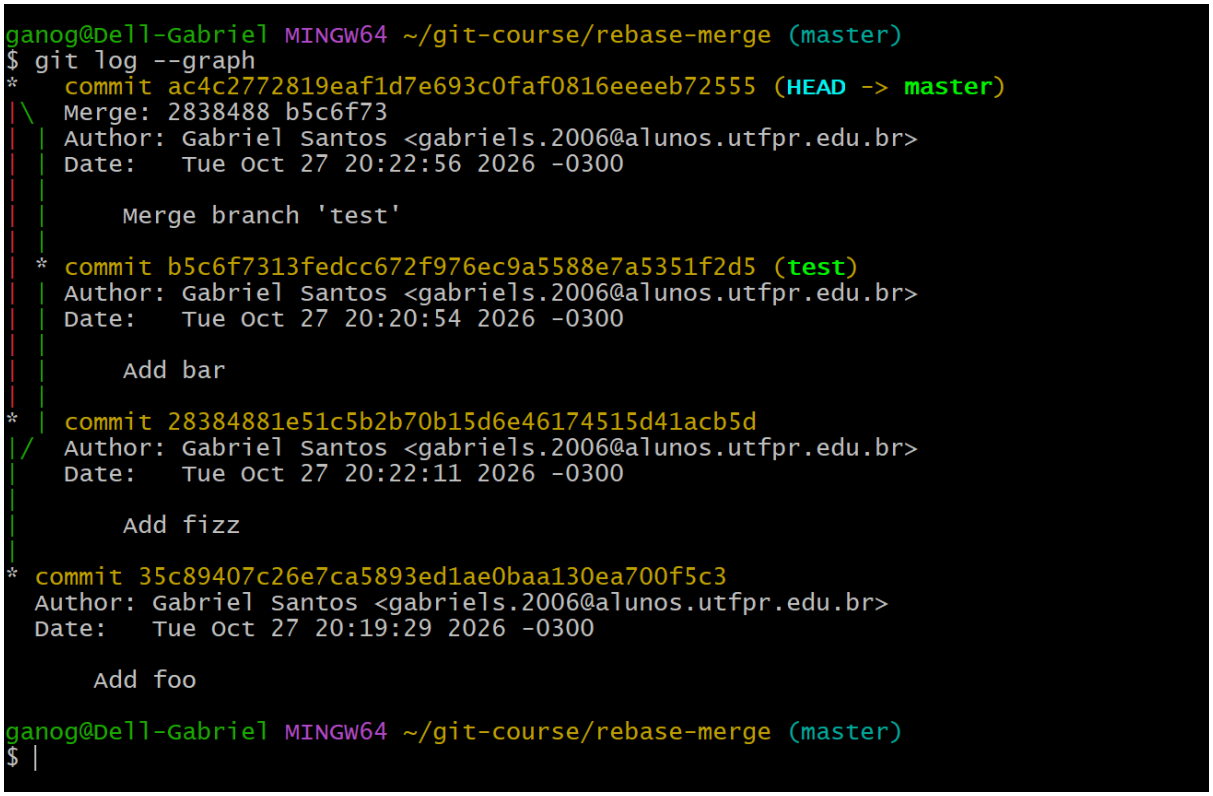
A outra forma é o Rebase, que ao invés de criar um novo commit, ele aproveita o ultimo commit para colocar no inicio da fila, o que pode deixar confuso a ordem dos commits, por não obedecer a ordem cronológica. Enquanto tá sempre atualizando, usar o rebase é indicado. Merge apenas para funções finais.

REBASE






MINGW64:/c/Users/ganog/git-course/rebase-merge
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (test)
$ git log
commit b5c6f7313fedcc672f976ec9a5588e7a5351f2d5 (HEAD -> test)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 20:20:54 2026 -0300
Add bar
commit 35c89407c26e7ca5893ed1ae0baa130ea700f5c3 (master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 20:19:29 2026 -0300
Add foo
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (test)
$ git checkout master
Switched to branch 'master'
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ git log
commit 35c89407c26e7ca5893ed1ae0baa130ea700f5c3 (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 20:19:29 2026 -0300
Add foo
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ vim fizz
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ git add fizz
warning: in the working copy of 'fizz', LF will be replaced by CRLF the next time Git touches it
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ git commit -m "Add fizz"
[master 2838488] Add fizz
1 file changed, 1 insertion(+)
create mode 100644 fizz
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ git log
commit 28384881e51c5b2b70b15d6e46174515d41acb5d (HEAD -> master)
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 20:22:11 2026 -0300
Add fizz
commit 35c89407c26e7ca5893ed1ae0baa130ea700f5c3
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 20:19:29 2026 -0300
Add foo
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$


ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ git log --graph
* commit ac4c2772819eaf1d7e693c0faf0816eeeeb72555 (HEAD -> master)
|\ Merge: 2838488 b5c6f73
| | Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
| | Date: Tue Oct 27 20:22:56 2026 -0300
| |
| | Merge branch 'test'
| |
| * commit b5c6f7313fedcc672f976ec9a5588e7a5351f2d5 (test)
| | Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
| | Date: Tue Oct 27 20:20:54 2026 -0300
| |
| | Add bar
| |
* | commit 28384881e51c5b2b70b15d6e46174515d41acb5d
|/ Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
| Date: Tue Oct 27 20:22:11 2026 -0300
|
| Add fizz
|
* commit 35c89407c26e7ca5893ed1ae0baa130ea700f5c3
Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
Date: Tue Oct 27 20:19:29 2026 -0300
Add foo
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$

```
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ git log --graph
* commit c4676b510b5463c28ef420c2fd4a1d040acbff8f (HEAD -> master)
| Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
| Date:   Tue Oct 27 20:28:02 2026 -0300
|
|     seila
|
* commit ac3b35b863266f15426f08093cd85b58ae62068d (rebase-branch)
| Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
| Date:   Tue Oct 27 20:26:43 2026 -0300
|
|     Add bla
|
* commit 6f443460b7a22a864232d76b42df0db03b3dbb81
| Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
| Date:   Tue Oct 27 20:25:33 2026 -0300
|
|     Add buzz
|
*   commit ac4c2772819eaf1d7e693c0faf0816eeeeb72555
|\  Merge: 2838488 b5c6f73
| | Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
| | Date:   Tue Oct 27 20:22:56 2026 -0300
| |
| |     Merge branch 'test'
| |
| * commit b5c6f7313fedcc672f976ec9a5588e7a5351f2d5 (test)
| | Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
| | Date:   Tue Oct 27 20:20:54 2026 -0300
| |
| |     Add bar
| |
* | commit 28384881e51c5b2b70b15d6e46174515d41acb5d
|/  Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
|   Date:   Tue Oct 27 20:22:11 2026 -0300
|
|       Add fizz
|
* commit 35c89407c26e7ca5893ed1ae0baa130ea700f5c3
  Author: Gabriel Santos <gabriels.2006@alunos.utfpr.edu.br>
  Date:   Tue Oct 27 20:19:29 2026 -0300

      Add foo

ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$
```

```
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ ls
bar  bla  buzz  fizz  foo  seila

ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ vi .gitignore

ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ git status
On branch master
Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .gitignore

nothing added to commit but untracked files present (use "git add" to track)

ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$ vi .gitignore

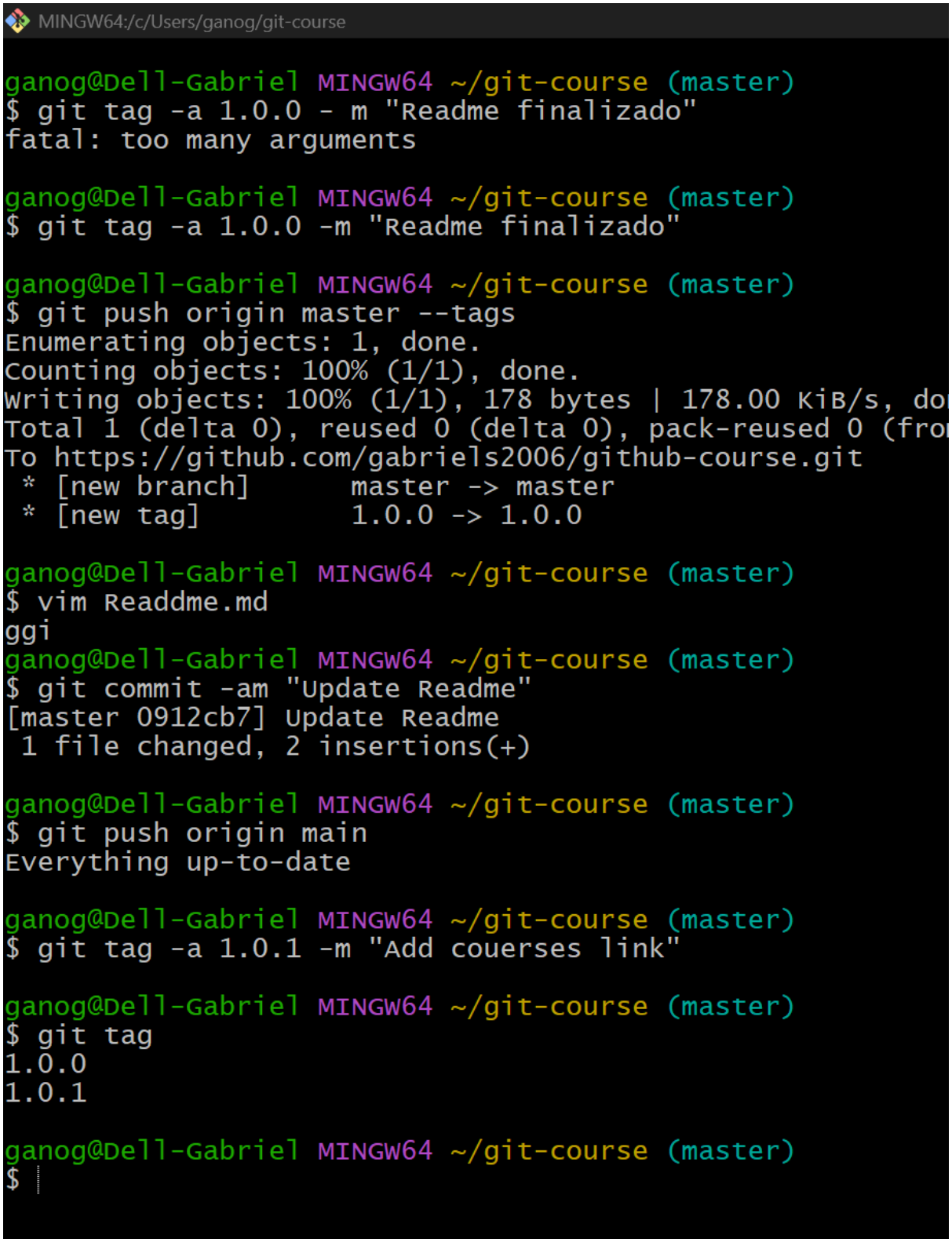
ganog@Dell-Gabriel MINGW64 ~/git-course/rebase-merge (master)
$
```

O stach salva sem mandar para o git direto, é ótimo para quando está mexendo ainda no projeto.

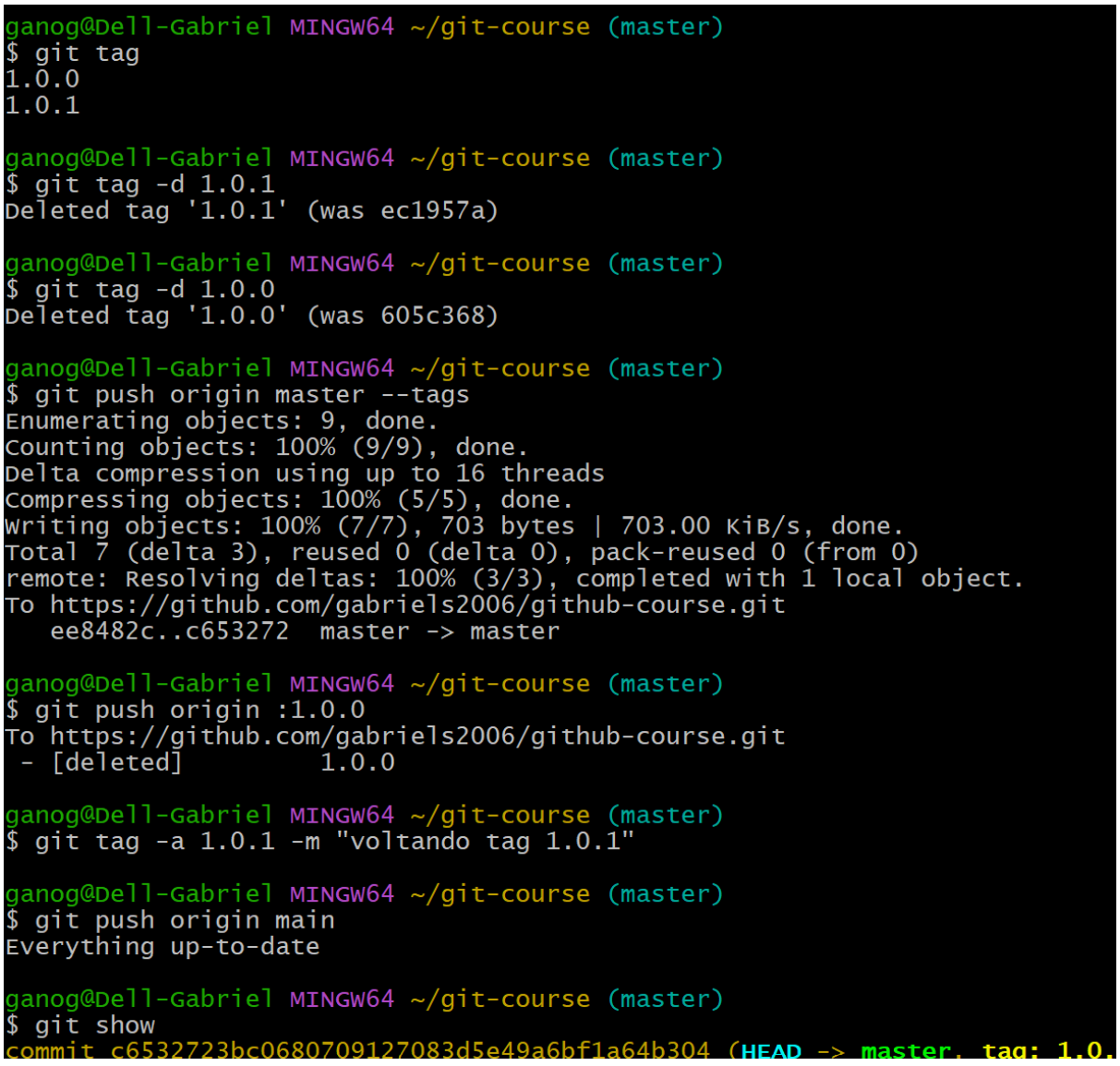
Alias também funciona para renomear funções, como o status mostrado na aula, renomeado para ‘s’.

Tags são funções muito úteis para deixar claro qual é a versão quando se trabalha com muitas versões.

LAMIA
Machine Learning for Industry







O revert vai ser útil pois não perde o que foi colocado, diferentemente do remove, que apaga tudo que foi commitado, nesse, o commit só reverte. Apagar tags funciona para quando a versão já não é mais bem vinda no repositório, por estar antiga demais ou a tag ter sido levantada errada.

3. Conclusão

O Git e GitHub resolvem problemas propondo uma solução simples, com poucos comandos para qualquer um compreender. Sua utilidade vai para além das ferramentas, sendo muito utilizadas por grandes corporações. Pude aprender comandos como o rebase que tem um principio parecido com o Merge, mas se diferencia por como o commit aparece de fato na linha e no histórico. Outros comandos também serão muito úteis para projetos futuros. Por fim, hoje Git e GitHub não são mais apenas uma escolha de versionamento, são a solução.